

MOCK CLINICAL TRIAL PROTOCOL SYNOPSIS

1. Study Identification

Study Title:

A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Parallel-Group Study to Evaluate the Efficacy and Safety of XYZ-101 in Patients with Type 2 Diabetes Mellitus

Study Number: XYZ-101-301

Investigational Product: XYZ-101

Study Phase: Phase 3

Study Population: Adult patients with Type 2 Diabetes Mellitus

Planned Number of Subjects: 120

Treatment Groups:

-Group 1: XYZ-101 — 60 subjects

- Group 2: Placebo — 60 subjects

2. Study Background

XYZ-101 is an investigational drug being evaluated for its efficacy and safety in patients with Type 2 Diabetes Mellitus.

This mock Phase 3 study is designed to evaluate whether XYZ-101 improves glycemic control compared with placebo while also assessing safety and tolerability.

Note: This is a simulated clinical trial created for Clinical Data Management (CDM) training purposes and does not represent an actual clinical study.

3. Study Objectives

3.1 Primary Objective

To evaluate the efficacy of XYZ-101 compared with placebo in improving glycemic control in patients with Type 2 Diabetes Mellitus.

3.2 Secondary Objectives

To evaluate:

- Change in fasting blood glucose from baseline.
- Change in body weight from baseline.
- Proportion of subjects achieving an HbA1c level of <7.0% at Day 112.
- Safety and tolerability of XYZ-101.

4. Study Design

This is a Phase 3, randomized, double-blind, placebo-controlled, parallel-group study.

A total of 120 eligible subjects will be randomized in a 1:1 ratio to receive either XYZ-101 or placebo.

Study Characteristics

Characteristic	Description

Phase	Phase 3
Study Type	Interventional
Design	Parallel-group
Blinding	Double-blind
Control	Placebo
Planned Subjects	120
Treatment Arms	2

5. Study Population

The study will include adult patients with Type 2 Diabetes Mellitus who meet all applicable inclusion criteria and none of the exclusion criteria.

5.1 Inclusion Criteria

Subjects must meet the following criteria:

1. Age 18–75 years.
2. Confirmed diagnosis of Type 2 Diabetes Mellitus.
3. HbA1c $\geq 7.0\%$ and $\leq 10.0\%$ at the Screening Visit.
4. BMI $\geq 18.5 \text{ kg/m}^2$ and $\leq 40.0 \text{ kg/m}^2$ at the Screening Visit.
5. Willing and able to comply with study procedures.
6. Able to provide informed consent.
7. Willing to attend all scheduled study visits.

5.2 Exclusion Criteria

Subjects meeting any of the following criteria will be excluded:

1. Diagnosis of Type 1 Diabetes Mellitus.

2. History of severe hypoglycemia within the protocol-defined period.
3. Significant uncontrolled cardiovascular disease.
4. Significant renal or hepatic impairment.
5. Known allergy or hypersensitivity to the investigational product.
6. Pregnancy or breastfeeding.
7. Participation in another clinical trial within the protocol-defined period.
8. Any medical condition that, in the investigator's opinion, may interfere with study participation or evaluation.

6. Treatment Arms

Treatment Arm 1

XYZ-101

Number of planned subjects: 60

Treatment Arm 2

Placebo

Number of planned subjects: 60

Subjects will be randomly assigned to treatment using the applicable randomization/IRT system.

7. Study Visits

Visit	Timing	Major Assessments
Screening	Day -14 to Day -1	Informed consent, eligibility, demographics, medical history, laboratory assessments

Visit 1-Randomization	Day 1	Eligibility confirmation, baseline assessments, randomization, treatment
Visit 2	Day 14	Vital signs, adverse events, treatment compliance
Visit 3	Day 28	Safety assessments, adverse events, laboratory assessments
Visit 4	Day 56	Efficacy assessments, safety assessments, laboratory assessments
Visit 5	Day 84	Efficacy assessments, safety assessments, laboratory assessments
End of Treatment	Day 112	Efficacy, safety and laboratory assessments
Follow-up	Day 126	Safety follow-up

8. Data to be Collected

The following major data domains will be collected during the study:

8.1 Demographics

- Subject ID
- Date of birth
- Age
- Sex
- Height
- Weight

8.2 Medical History

- Medical condition
- Start date
- End date
- Ongoing status

8.3 Concomitant Medications

- Medication name

- Dose
- Route
- Frequency
- Start date
- End date
- Indication
- Ongoing status

8.4 Vital Signs

- Systolic blood pressure
- Diastolic blood pressure
- Pulse
- Respiratory rate
- Temperature
- Weight

8.5 Laboratory Assessments

Hematology

- Hemoglobin
- WBC
- Platelets

Clinical Chemistry

- ALT
- AST
- Creatinine
- Glucose

Diabetes-related assessments

- HbA1c

- Fasting blood glucose

8.6 Adverse Events

The following information will be collected for adverse events:

- AE term
- Start date
- End date
- Severity
- Seriousness
- Relationship to study treatment
- Action taken
- Outcome

8.7 Treatment Exposure

- Treatment received
- Dose
- Administration date
- Number of doses
- Treatment interruption
- Treatment discontinuation

9. Study Endpoints

9.1 Primary Endpoint

Change from baseline in HbA1c at Day 112.

9.2 Secondary Endpoints

1. Change from baseline in fasting blood glucose at Day 112.
2. Change from baseline in body weight at Day 112.
3. Proportion of subjects with an HbA1c level of <7.0% at Day 112.

4. Incidence of treatment-emergent adverse events.

10. Safety Assessments

Safety will be evaluated through:

- Adverse event monitoring.
- Serious adverse event monitoring.
- Vital sign assessments.
- Laboratory assessments.
- Concomitant medication review.
- Treatment exposure and treatment discontinuation information.

11. Role of Clinical Data Management

The clinical/study team develops the clinical protocol; Clinical Data Management does not create it.

For this CDM training project, the mock protocol serves as the source document from which CDM requirements will be derived.

CDM activities will include:

- Reviewing protocol data requirements.
- Identifying data that need to be collected.
- Supporting CRF/eCRF design.
- Applying appropriate data standards.
- Developing data-management specifications.
- Defining edit checks.
- Supporting database/EDC build.
- Performing UAT.
- Managing queries and data discrepancies.
- Supporting external data reconciliation.

- Performing data review and cleaning.
- Supporting database freeze and database lock.

12. Purpose of This Protocol Synopsis

This mock protocol synopsis will serve as the reference document for the subsequent stages of the CDM project.

The information defined here will be used to develop:

Protocol → CRF/eCRF → CDASH considerations → Annotated eCRF → EDC/database → Edit Checks → UAT → Data Entry → Query Management → Data Cleaning → Database Lock

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